

XTSF: Synthetic Forecasting Explanations with XPC

Experiment Guideline

Thesis project

7 August 2026

1 Research question

XTSF asks whether Monte Carlo Shapley explanations recover known contributions in a controlled forecasting problem, and distinguishes:

1. estimator error relative to the chosen exact Shapley estimand;
2. estimand mismatch relative to the known structural decomposition.

This prevents convergence to a well-defined but scientifically inappropriate estimand from being mistaken for recovery of ground truth.

2 Synthetic structural ground truth

The hourly response is a sum of explicit weather, calendar, trend, interaction, and irrelevant components. Each feature group therefore has a known signed structural contribution. The trained forecasting model consumes lagged targets and exogenous features through a common tabular adapter. Generator parameters, noise, train/test split, and model fitting must all be fixed by one run seed.

3 Shapley estimands

For feature groups $N = \{1, \dots, n\}$ and coalition value $v(S)$,

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)].$$

The interventional value masks absent features independently from their observed dependence. The conditional value uses nearest-neighbor conditional backgrounds. Exact enumeration defines each method’s estimand; Monte Carlo coalition and mask sampling approximate it.

4 Convergence experiment

The two sampling axes are varied separately over $\{1, 2, 4, 8, 16, 32\}$:

- sampled coalitions with 16 masking samples fixed;
- masking samples with 16 sampled coalitions fixed.

Each configuration is repeated across fixed seeds. Report MSE against both the exact method-specific estimand and the signed structural contribution. The interventional estimand equals the structural target in the current additive construction; the conditional estimand need not because dependence redistributes credit.

5 Aggregation games

Three ways to explain grouped forecasts are compared:

Mode	Game	Practical trade-off
default	explain fine features, then aggregate attributions	largest model-row count
coalitional	define groups as players before Shapley estimation	aligned grouped game, moderate cost
simplified	two-player grouped approximation	fastest, changed estimand and coarser interaction allocation

Report structural MSE, exact-estimand MSE, efficiency RMSE, time per explained point, and model rows per point.

6 Implementation

Reusable code lives in `src/xpc/`: data/model adapters, maskers, feature groups, coalition values, exact/Monte Carlo estimators, heightening, and aggregation helpers. The maintained notebook `src/xtsf.ipynb` defines the reference experiment and exports tables and figures.

The synthetic panel cache is `outputs/synthetic_forecasting_panel.csv`. Set `REBUILD_SYNTHETIC=True` whenever generator settings or the seed change; otherwise the existing fixed file is deliberately reused.

7 Artifacts

The notebook writes convergence run/summary CSVs and a plot, exact aggregation estimands, aggregation run/summary CSVs and a plot, plus JSON plot data. Output filenames are fixed. A rerun replaces them, so distinct configurations require distinct output roots or complete regeneration of all related artifacts.

8 Validation and interpretation

Run the package unit tests and the notebook smoke in the prepared environment. Verify efficiency, grouping consistency, exact-versus-Monte-Carlo agreement, and signed structural scale before interpreting plots. Conclusions about conditional versus interventional explanations are specific to the chosen background distribution and structural target.